

DESI DR2 BAO measurements — verified from source (Cobaya **bao_data/desi_bao_dr2, ALL_GCcomb**)

Provenance: pulled from github.com/CobayaSampler/bao_data/desi_bao_dr2/desi_gaussian_bao_ALL_GCcomb_{mean,cov} — the machine-readable measurement + covariance files the DESI/Cobaya pipeline uses. NOT remembered numbers, NOT the CPL/ w_0w_a fit (which would be circular for the discriminator). Keeper pulled + parsed 2026-08-13 (K1436). This is the data Grace’s radial-D_H monotone-vs-flip discriminator (grace_A1, 2026-08-08) was waiting on.

Full table (D/r_d, with 1σ)

z_eff	tracer	D_M/r_d	D_H/r_d	D_V/r_d	r(D_M,D_H)
0.295	BGS	—	—	7.9417 ± 0.0761	(isotropic)
0.510	LRG1	13.5876 ± 0.1684	21.8629 ± 0.4289	—	−0.452
0.706	LRG2	17.3507 ± 0.1799	19.4553 ± 0.3339	—	−0.395
0.934	LRG3+ELG1	21.5756 ± 0.1618	17.6415 ± 0.2010	—	−0.347
1.321	ELG2	27.6009 ± 0.3246	14.1760 ± 0.2246	—	−0.398
1.484	QSO	30.5119 ± 0.7636	12.8170 ± 0.5180	—	−0.494
2.330	Ly α	38.9890 ± 0.5317	8.6315 ± 0.1011	—	−0.431

★ The radial series the discriminator uses (D_H/r_d vs z, model-independent — no w(a) assumed)

z=0.510 D_H/r_d = 21.863 ± 0.429
z=0.706 D_H/r_d = 19.455 ± 0.334
z=0.934 D_H/r_d = 17.641 ± 0.201
z=1.321 D_H/r_d = 14.176 ± 0.225
z=1.484 D_H/r_d = 12.817 ± 0.518
z=2.330 D_H/r_d = 8.632 ± 0.101

(BGS z=0.295 gives only the isotropic D_V — no D_H point.)

★ Covariance structure (a simplification for Grace)

The ALL_GCcomb covariance is BLOCK-DIAGONAL: each z-bin is statistically independent of every other bin (all cross-bin entries are exactly 0). The only correlations are D_M $\leftrightarrow$ D_H *within* a bin ($r \approx -0.35$ to -0.49 , tabulated above). Consequences for the discriminator: - The six D_H points are mutually independent $\rightarrow$ for a pure radial-D_H monotonicity test you can use the diagonal D_H errors directly; no cross-bin covariance to carry. - The D_M–D_H anti-correlation only matters if you marginalize D_M or build R(z) from a joint (D_M, D_H) reference. If the discriminator is D_H-only, it’s clean diagonal. - Raw covariance (13×13, file order: DV@0.295; then DM,DH per bin at 0.510/0.706/0.934/1.321/1.484; then DH,DM at 2.330) is in the Cobaya file; the per-bin 2×2 blocks are recoverable from (σ_{DM} , σ_{DH} , r) above.

What this unblocks

Grace’s discriminator (grace_A1): BST’s completely-monotone w(a) $\rightarrow$ monotone radial residual R(z) for every amplitude; the CPL phantom fit $\rightarrow$ residual flips sign near $z \approx 0.9$. With this table the test can now fire: compute R(z) against a fiducial r_d/H₀ reference, test monotonicity vs. the flip, report the significance. Do not substitute the CPL best-fit as “data” (circular — Grace’s own guard). This table is the model-independent input; the verdict is Grace’s to compute.

— assembled by Keeper, K1436, 2026-08-13, verified at source.